

Linux Commands Cheat Sheet

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The Linux Terminal

Getting Help in Linux

MAN Pages

- Use `man command` to view the manual for a command.
 - Example: `man ls`

The man pages are navigated using the `less` command with shortcuts:

- `h`: Get help within `less`.
- `q`: Quit `less`.
- `enter`: Show next line.
- `space`: Show next screen.
- `/string`: Search forward for a string.
- `?string`: Search backward for a string.
- `n` / `N`: Next/previous appearance of the search term.

Checking Command Type

- `type rm`: Check if `rm` is a shell built-in or an executable file.
 - Example: `rm is /usr/bin/rm`
- `type cd`: Check if `cd` is a shell built-in.
 - Example: `cd is a shell builtin`

Getting Help for Shell Built-in Commands

- `help command`: Get help for shell built-in commands.
 - Example: `help cd`
- `command --help`: Get help for executable commands.
 - Example: `rm --help`

Searching Man Pages

- `man -k uname`: Search for `uname` in all man pages.
- `man -k "copy files"`: Search for "copy files" in man pages.
- `apropos passwd`: Search for `passwd` related man pages.

Keyboard Shortcuts

```
TAB TAB: Display all commands or filenames starting with written letters.
CTRL + L: Clear the current line.
CTRL + D: Close the shell.
CTRL + U: Cut the current line.
CTRL + A: Move cursor to start of the line.
Ctrl + E: Move cursor to the end of the line.
CTRL + C: Stop the current command.
CTRL + Z: Sleep the running program.
CTRL + ALT + T: Open a terminal.
```

Bash History

- `history`: Display the history.
- `history -d 100`: Remove a specific line from history.
- `history -c`: Clear the entire history.
- `echo $HISTFILESIZE`: Print the number of commands saved in the history file.
- `echo $HISTSIZE`: Print the number of history commands saved in memory.
- `!!`: Rerun the last command.
- `!20`: Run a specific command from history.
- `!-10`: Run the last nth command.
- `!abc`: Run the last command starting with `abc`.
- `!abc:p`: Print the last command starting with `abc`.
- `CTRL + R`: Reverse search through history.
- `HISTTIMEFORMAT="%d/%m/%y %T "`: Record date/time of each command (add to `~/.bashrc` for persistence).

Getting Root Access (sudo, su)

- `sudo command`: Run a command as root (for users in `sudo` or `wheel` group).
- `sudo su`: Become root temporarily.
- `sudo passwd root`: Set the root password.
- `passwd username`: Change a user's password.
- `su`: Become root (if root has a password).

Linux Paths

Paths

- **Absolute**: Starts with `/`
- **Relative**: Relative to the current location
- `.`: Current working directory
- `..`: Parent directory
- `~`: User's home directory

Changing Directories

- `cd`: To user's home directory

- `cd ~`: To user's home directory
- `cd -`: To the last directory
- `cd /path_to_dir`: To `path_to_dir`
- `pwd`: Print the current working directory

Installing Tools

- `sudo apt install tree`: Installs the `tree` command

Using Tree

- `tree directory/`: Example: `tree .`
- `tree -d .`: Print only directories
- `tree -f .`: Print absolute paths

The ls Command

Usage: `ls [OPTIONS] [FILES]`

Listing Directories

- `ls`, `ls .`: Current directory
- `ls ~ /var /`: Multiple directories

Options

- `-l`: Long listing
- `-a`: All files (including hidden)
- `-1`: Single column
- `-d`: Directory information
- `-h`: Human-readable sizes
- `-S`: Sort by size
- `-X`: Sort by extension
- `--hide`: Hide specific files
- `-R`: Recursive listing
- `-i`: Inode number

Disk Usage

- `du -sh ~`: Size of home directory

File Timestamps and Date

- `ls -lu`: Access time (`atime`)
- `ls -l`, `ls -lt`: Modification time (`mtime`)
- `ls -lc`: Change time (`ctime`)

- `stat file.txt`: All timestamps
- `ls -l --full-time /etc/`: Full timestamps

Modifying Timestamps with Touch

- `touch file.txt`: Create or update timestamps
- `touch -a file`, `touch -m file`: Modify `atime` or `mtime`
- `touch -m -t 201812301530.45 a.txt`: Specific date/time
- `touch -d "2010-10-31 15:45:30" a.txt`: Both `atime` and `mtime`
- `touch a.txt -r b.txt`: Copy timestamps

Date and Calendar

- `date`: Current date/time
- `cal`, `cal 2021`, `cal 7 2021`: Calendars
- `cal -3`: Previous, current, next month
- `date --set="2 OCT 2020 18:00:00"`: Set date/time

Sorting with ls

- `ls -l`: Sorted by name
- `ls -lt`: Sorted by `mtime`, newest first
- `ls -ltu`: Sorted by `atime`
- `ls -ltu --reverse`: Reverse order

Viewing Files (cat, less, more, head, tail, watch)

Displaying File Contents

- `cat filename`: Display content
- `cat -n filename`: Line numbers
- `cat filename1 filename2 > filename3`: Concatenate

Less Shortcuts

- `h`: Help
- `q`: Quit
- `enter`: Next line
- `space`: Next screen
- `/string`: Search forward
- `?string`: Search backward
- `n` / `N`: Next/previous search result

Tail and Head

- `tail filename`: Last 10 lines

- `tail -n 15 filename`: Last 15 lines
- `tail -n +5 filename`: Starting with line 5
- `tail -f filename`: Real-time updates
- `head filename`: First 10 lines
- `head -n 15 filename`: First 15 lines

Monitoring Commands

- `watch -n 3 ls -l`: Refresh every 3 seconds

Working with Files and Directories

Creating and Updating Files

- `touch filename`: Create a new file or update timestamps.

Creating Directories

- `mkdir dir1`: Create a new directory.
- `mkdir -p mydir1/mydir2/mydir3`: Create nested directories.

The cp Command

Copy files and directories:

- `cp file1 file2`: Copy `file1` to `file2`.
- `cp file1 dir1/file2`: Copy to another directory with a different name.
- `cp -i file1 file2`: Prompt before overwrite.
- `cp -p file1 file2`: Preserve permissions.
- `cp -v file1 file2`: Verbose output.
- `cp -r dir1 dir2/`: Recursively copy directories.
- `cp -r file1 file2 dir1 dir2 dest_dir/`: Copy multiple items to a destination.

The mv Command

Move or rename files and directories:

- `mv file1 file2`: Rename a file.
- `mv file1 dir1/`: Move to a directory.
- `mv -i file1 dir1/`: Prompt before overwrite.
- `mv -n file1 dir1/`: Prevent overwriting.
- `mv -u file1 dir1/`: Update based on modification time.
- `mv file1 dir1/file2`: Move and rename.
- `mv file1 file2 dir1/ dir2/ dest_dir/`: Move multiple items.

The rm Command

Remove files and directories:

- `rm file1`: Remove a file.
- `rm -v file1`: Verbose removal.
- `rm -r dir1/`: Remove a directory.
- `rm -rf dir1/`: Force removal without prompt.
- `rm -ri file1 dir1/`: Prompt for each removal.

Secure File Deletion

- `shred -vu -n 100 file1`: Securely overwrite and remove a file.

Piping and Command Redirection

Piping Examples

- `ls -lSh /etc/ | head`: View the top 10 largest files.
- `ps -ef | grep sshd`: Check if `sshd` is running.
- `ps aux --sort=-%mem | head -n 3`: Top 3 processes by memory.

Command Redirection

Redirect output and errors:

- `ps aux > processes.txt`: Output to a file.
- `id >> users.txt`: Append output.
- `tail -n 10 /var/log/*.log > output.txt 2> errors.txt`: Separate output and errors.
- `tail -n 2 /etc/passwd /etc/shadow > all.txt 2>&1`: Redirect all to one file.
- `cat /var/log/auth.log | grep "fail" | wc -l`: Count occurrences.

Finding Files with locate and find

locate

- `sudo apt install plocate`: Install `plocate`.
- `sudo updatedb`: Update the database.
- `locate filename`: Find a file by name.
- `locate -i filename`: Case insensitive search.
- `locate -b '\filename'`: Exact name search.
- `locate -r 'regex'`: Regular expression search.
- `locate -e filename`: Check file existence.
- `which command`: Show command path.

find

Search with various options:

- `find ~ -type f -size +1M`: Files over 1MB.
- Options include `-type`, `-name`, `-iname`, `-size`, `-perm`, `-links`, `-atime`, `-mtime`, `-ctime`, `-user`, and `-group`.

Searching for Text Patterns with grep

Usage: `grep [OPTIONS] PATTERN FILE`

Options

- `-n`: Print line number.
- `-i`: Case insensitive.
- `-v`: Invert match.
- `-w`: Match whole words.
- `-a`: Include binary files.
- `-R`: Recursive search.
- `-c`: Count matches.
- `-C n`: Context display (n lines around the match).

Extracting ASCII Characters from Binary Files

- `strings binary_file`: Example `strings /bin/ls`.

VIM – Text Editor

Modes

- Command Mode: Default on entry.
- Insert Mode: Editing text.
- Last Line Mode: Save/exit commands.

Config File

- VIM settings: `~/.vimrc`.

Commands

- `i`, `I`, `a`, `A`, `O`: Enter Insert Mode.
- `:w!`, `:q!`, `:wq!`, `:e!`: Save/quit commands in Last Line Mode.
- `x`, `dd`, `ZZ`, `u`, `G`, `$`, `O`, `^`: Editing commands in Command Mode.
- `/string`, `?string`, `n`, `N`: Search commands in Command Mode.
- `vim -o file1 file2`: Open files in stacked windows.
- `vim -d file1 file2`: Highlight differences.

Navigation

- `Ctrl+W`: Switch between files.

Account Management

```
## Account Management
/etc/passwd # users and info:
/etc/shadow # users' passwords
/etc/group # groups

## User Commands
useradd [OPTIONS] username # Create user.
usermod [OPTIONS] username # Modify user.
userdel -r username        # Delete user.

## Group Commands
groupadd group_name # Create group.
groupdel group_name # Delete group.

## Examples
useradd -m -d /home/john -c "C++ Developer" -s /bin/bash -G sudo,adm,mail john # Example of creating a user.
usermod -aG developers,managers john # Example of modifying a user.
```

Monitoring Users

```
## Commands
who -H      # User info.
id          # User info.
whoami      # User info.
w           # System usage.
uptime     # System usage.
last        # Login history.
last -u username # Login history for a specific user.
```

File Permissions

Understanding Permissions

- **Legend:** `u` (user), `g` (group), `o` (others), `a` (all), `r` (read), `w` (write), `x` (execute), `-` (no access).

Displaying Permissions

- `ls -l /etc/passwd`: View file permissions.
- `stat /etc/shadow`: Detailed permission stats.

Changing Permissions

- `chmod u+r filename`: Add read to user.
- `chmod u+r,g-wx,o-rwx filename`: Adjust multiple permissions.
- `chmod ug+rwx,o-wx filename`: Set multiple permissions.
- `chmod ugo+x filename`: Add execute to all.

- `chmod a+r,a-wx filename`: Modify all permissions.

Absolute Mode

- `chmod 777 filename`: Set all permissions for all.
- `chmod 755 filename`: Read & execute for group and others.
- `chmod 644 filename`: Read-only for group and others.

Special Permissions

- SUID: `chmod u+s executable_file`.
- SGID: `chmod g+s projects/`.
- Sticky Bit: `chmod o+t temp/`.

UMASK

- Display: `umask`.
- Set new value: `umask new_value`.

Ownership

- Owner: `chown new_owner file`.
- Group: `chgrp new_group file`.
- Both: `chown new_owner:new_group file`.
- Recursive: `chown -R new_owner file`.

File Attributes

- Display: `lsattr filename`.
- Change: `chattr +-attribute filename`.

Processes

Process Viewing

- `type rm`: Check if `rm` is built-in or executable.
- `ps`: Processes in current terminal.
- `ps -ef`, `ps aux`, `ps aux | less`: System processes.
- `ps aux --sort=%mem | less`: Sort by memory usage.
- `ps -ef --forest`: ASCII process tree.
- `ps -f -u username`: Processes by user.
- `pgrep -l sshd`, `pgrep -f sshd`, `ps -ef | grep sshd`: Check for `sshd`.
- `pstree`, `pstree -c`: Hierarchical process tree.

Dynamic Real-Time View

- `top`: Start system monitor.

- `top` shortcuts: `h` for help, `space` for refresh, `d` for delay, etc.
- `top -d 1 -n 3 -b > top_processes.txt`: Top in batch mode.
- Install `htop` for an interactive view.

Killing Processes

- `kill -l`: List signals.
- `kill pid`, `kill -SIGNAL pid1 pid2 ...`: Send signals.
- `kill -2 pid`, `kill -HUP pid`: Send specific signal.
- `pkill process_name`, `killall process_name`: Kill by name.
- `kill $(pidof process_name)`: Kill using `pidof`.

Background and Foreground Management

- `command &`: Run in background.
- `jobs`: List jobs.
- `Ctrl + Z`: Stop process.
- `fg %job_id`: Resume in foreground.
- `bg %job_id`: Resume in background.
- `nohup command &`: Immune to hangups.

Networking

Getting Network Interface Information

- `ifconfig`: Enabled interfaces.
- `ifconfig -a`, `ip address show`: All interfaces.
- `ifconfig enp0s3`, `ip addr show dev enp0s3`: Specific interface.
- `ip -4 address`: Only IPv4 info.
- `ip -6 address`: Only IPv6 info.
- `ip link show`, `ip link show dev enp0s3`: L2 info, including MAC.
- `route`, `route -n`, `ip route show`: Default gateway.
- `systemd-resolve --status`: DNS servers.

Setting Network Interfaces

- `ifconfig enp0s3 down`, `ip link set enp0s3 down`: Disable interface.
- `ifconfig enp0s3 up`, `ip link set enp0s3 up`: Enable interface.
- `ifconfig -a`, `ip link show dev enp0s3`: Check status.
- `ifconfig enp0s3 192.168.0.222/24 up`, `ip address add 192.168.0.112/24 dev enp0s3`: Set IP.
- `ifconfig enp0s3:1 10.0.0.1/24`: Secondary IP.
- `route del default gw`, `ip route del default`, `ip route add default via`: Default gateway.
- `ifconfig enp0s3 hw ether`, `ip link set dev enp0s3 address`: Change MAC.

Netplan for Static Network Configuration on Ubuntu

1. Stop/Disable NetworkManager.
2. Create/Modify YAML in `/etc/netplan`.
3. Apply config: `sudo netplan apply`.
4. Verify: `ifconfig`, `route -n`.

OpenSSH Configuration and Management

Installation

Ubuntu

```
sudo apt update && sudo apt install openssh-server openssh-client
```

CentOS

```
sudo dnf install openssh-server openssh-clients
```

Server Connection

```
ssh -p 22 username@server_ip # Connect using default SSH port
ssh -p 22 -l username server_ip # Connect with a specific username
ssh -v -p 22 username@server_ip # Connect in verbose mode for detailed information

# Ubuntu
sudo systemctl status ssh # Check SSH status
sudo systemctl stop ssh # Stop SSH service
sudo systemctl restart ssh # Restart SSH service
sudo systemctl enable ssh # Enable SSH to start on boot
sudo systemctl is-enabled ssh # Check if SSH is enabled on boot

# CentOS
sudo systemctl status sshd # Check SSH status
sudo systemctl stop sshd # Stop SSH service
sudo systemctl restart sshd # Restart SSH service
sudo systemctl enable sshd # Enable SSH to start on boot
sudo systemctl is-enabled sshd # Check if SSH is enabled on boot
```

Security Configuration

Edit `/etc/ssh/sshd_config` and then apply changes by restarting SSH:

- Change port: Port 2278
- Disable root login: `PermitRootLogin no`
- Restrict user access: `AllowUsers user1 user2`
- Configure firewall to filter SSH access
- Enable Public Key Authentication, disable password-based login
- Use SSH Protocol 2 only
- Set client session intervals and max attempts for security

Remember to consult the man page (`man sshd_config`) for detailed configuration options.

File Transfer Techniques with SCP and RSYNC

SCP Usage

```
# Copy local file to remote host
scp a.txt john@80.0.0.1:~
scp -P 2288 a.txt john@80.0.0.1:~ # Custom port

# Copy from remote to local
scp -P 2290 john@80.0.0.1:~/a.txt .

# Copy entire directory to remote
scp -P 2290 -r projects/ john@80.0.0.1:~
```

RSYNC Commands

```
# Sync local directory to local backup
sudo rsync -av /etc/ ~/etc-backup/

# Mirror directory, deleting extraneous files from dest
sudo rsync -av --delete /etc/ ~/etc-backup/

# Exclude files during sync
rsync -av --exclude-from='~/exclude.txt' /source/ /dest/

# Sync over SSH with custom port
sudo rsync -av -e 'ssh -p 2267' /etc/ student@192.168.0.108:~/etc-backup/
```

Exclude Patterns Example

```
# exclude.txt could include patterns like:
*.avi
music/
abc.mkv

# Exclude specific file types during transfer
rsync -av --exclude='*.mkv' /source/ /dest/
```

WGET for File Download

```
# Install wget
sudo apt install wget # Ubuntu
sudo dnf install wget # CentOS

# Basic file download
wget https://example.com/file.iso

# Resume incomplete download
wget -c https://example.com/file.iso

# Download with bandwidth limit
wget --limit-rate=100k https://example.com/file.iso

# Download multiple files
wget -i urls.txt # urls.txt contains list of URLs
```

```
# Recursive download for offline viewing of a website
wget -mkEpng http://example.org
```

Use these commands to efficiently copy files and directories across systems and for downloading content from the internet, ensuring data synchronization and maintaining web accessibility.

NETSTAT and SS Usage

```
# Display all ports and connections
sudo netstat -tupan
sudo ss -tupan

# Check if port 80 is open
netstat -tupan | grep :80
```

LSOF Commands

```
# List open files
lsof

# Files opened by a specific user
lsof -u username

# Files opened by a specific command/process
lsof -c sshd

Open files for TCP ports in LISTEN state
lsof -iTCP -sTCP:LISTEN
lsof -iTCP -sTCP:LISTEN -nP
```

Use these commands to monitor network connections, check for open ports, and view files opened by users or processes, especially for security and troubleshooting.

Nmap Scanning Guide

```
# SYN Scan (root required)
nmap -sS 192.168.0.1

# TCP Connect Scan
nmap -sT 192.168.0.1

# Scan All Ports
nmap -p- 192.168.0.1

# Scan Specific Ports
nmap -p 20,22-100,443,1000-2000 192.168.0.1

# Service Version Detection
nmap -p 22,80 -sV 192.168.0.1

# Ping Scan Network
nmap -sP 192.168.0.0/24
```

```
# Skip Host Discovery
nmap -Pn 192.168.0.0/24

# Exclude Specific IP from Scan
nmap -sS 192.168.0.0/24 --exclude 192.168.0.10

# Output Scan to File
nmap -oN output.txt 192.168.0.1

# OS Detection
nmap -O 192.168.0.1

# Aggressive Scan
nmap -A 192.168.0.1

# Read Targets from File & Output to File without DNS Resolution
nmap -n -iL hosts.txt -p 80 -oN output.txt
```

Only scan your own networks and systems, or those you have explicit permission to test. Unauthorized scanning can be illegal.

Software Management with DPKG and APT

DPKG

- View .deb file info: `dpkg --info package.deb`
- Install from .deb: `sudo dpkg -i package.deb`
- List installed programs: `dpkg --get-selections` or `dpkg-query -l`
- Find by name: `dpkg-query -l | grep ssh`
- List package files: `dpkg -L openssh-server`
- Find owning package: `dpkg -S /bin/lis`
- Remove package: `sudo dpkg -r package`
- Purge package: `sudo dpkg -P package`

APT

- Update index: `sudo apt update`
- Install/update: `sudo apt install apache2`
- List upgradable: `sudo apt list --upgradable`
- Full upgrade: `sudo apt full-upgrade`
- Remove: `sudo apt remove package`
- Purge: `sudo apt purge package`
- Auto remove dependencies: `sudo apt autoremove`
- Clean cache: `sudo apt clean`
- List all packages: `sudo apt list`
- Search: `sudo apt list | grep nginx`
- Show package info: `sudo apt show nginx`
- List installed: `sudo apt list --installed`

Task Scheduling using Cron

```
crontab -e # Edit crontab
crontab -l # List tasks
crontab -r # Remove tasks

# Schedule Format:
* * * * * command # Every minute
15 * * * * command # Hourly
30 18 * * * command # Daily
3 22 * * 1 command # Weekly
10 6 1 * * command # Monthly
@yearly command # Yearly
@reboot command # At reboot
```

Getting System Hardware Information

General Hardware

```
lshw # Full hardware info
lshw -short # Short format
lshw -json # JSON format
lshw -html # HTML format
```

CPU Information

```
lscpu # CPU details
lshw -C cpu # Hardware-specific CPU details
lscpu -J # JSON format
```

Memory Information

```
dmidecode -t memory # RAM specs
dmidecode -t memory | grep -i size
dmidecode -t memory | grep -i max
free -m # Memory usage
```

PCI and USB Devices

```
lspci # PCI buses and connected devices
lspci | grep -i wireless
lspci | grep -i vga
lsusb # USB controllers and devices
lsusb -v # Verbose output
```

Storage Devices

```
lshw -short -C disk
fdisk -l # List disks
fdisk -l /dev/sda
lsblk # Block devices list
```


Network Devices

```
lshw -C network
iw list           # Wi-Fi cards
iwconfig          # Wi-Fi configuration
iwlist scan       # Wi-Fi networks scan
```

System Information via /proc

```
cat /proc/cpuinfo      # CPU info
cat /proc/meminfo      # Memory info
cat /proc/version      # System version
uname -r               # Kernel version
uname -a               # All system info
```

Battery Power

```
acpi -bi             # Battery info
acpi -V              # All ACPI info
```

Working with Device Files (dd)

```
# Backup MBR
dd if=/dev/sda of=~ /mbr.dat bs=512 count=1

# Restore MBR
dd if=~ /mbr.dat of=/dev/sda bs=512 count=1

# Clone partition
dd if=/dev/sda1 of=/dev/sdb2 bs=4M status=progress
```

Use these commands to check hardware specifications and perform operations with device files safely.

Service Management

```
# Analyze boot process
systemd-analyze
systemd-analyze blame

# List active units
systemctl list-units
systemctl list-units | grep ssh

# Service status
sudo systemctl status nginx.service

# Stop service
sudo systemctl stop nginx

# Start service
sudo systemctl start nginx

# Restart service
sudo systemctl restart nginx
```

```
# Reload service config
sudo systemctl reload nginx
sudo systemctl reload-or-restart nginx

# Enable service at boot
sudo systemctl enable nginx

# Disable service at boot
sudo systemctl disable nginx

# Check if service is enabled at boot
sudo systemctl is-enabled nginx

# Mask service
sudo systemctl mask nginx

# Unmask service
sudo systemctl unmask nginx
```

Ubuntu

```
sudo systemctl status ssh      # Check SSH service status
sudo systemctl stop ssh        # Stop SSH service
sudo systemctl restart ssh     # Restart SSH service
sudo systemctl enable ssh      # Enable SSH to start on boot
sudo systemctl is-enabled ssh  # Check if SSH is enabled on boot
```

CentOS

```
sudo systemctl status sshd     # Check SSHD service status
sudo systemctl stop sshd       # Stop SSHD service
sudo systemctl restart sshd    # Restart SSHD service
sudo systemctl enable sshd     # Enable SSHD to start on boot
sudo systemctl is-enabled sshd # Check if SSHD is enabled on boot
```

Security Configuration

To configure security settings, edit `/etc/ssh/sshd_config`. Apply changes by restarting SSH. Key configurations include:

- Change SSH port:
 - `Port 2278`
- Disable root login:
 - `PermitRootLogin no`
- Restrict user access to specified users only:
 - `AllowUsers user1 user2`
- Configure firewall to filter SSH access
- Enable Public Key Authentication and disable password-based login
- Use SSH Protocol 2 only
- Set client session intervals and maximum attempts for increased security

Note: Consult the man page (`man sshd_config`) for detailed configuration options.

Bash Programming

Bash Aliases

```
alias                # List all aliases
alias name='command' # Create an alias
unalias name         # Remove an alias
```

Useful Aliases

```
alias c='clear'
alias cl='clear; ls; pwd'
alias root='sudo su'
alias ports='netstat -tupan'
alias sshconfig='sudo vim /etc/ssh/sshd_config'
alias update='sudo apt update && sudo apt dist-upgrade -y && sudo apt clean'
```

Interactive File Manipulation

```
alias cp='cp -i'
alias mv='mv -i'
alias rm='rm -i'
```

Bash Variables

```
variable="value"      # Define a variable
echo $variable         # Reference a variable
declare -r const=100  # Define a read-only variable
unset variable         # Unset a variable
env | grep PATH        # Find an environment variable
export PATH=$PATH:~/bin # Modify the PATH variable
```

Special Variables

```
$0, $1, $2, ..., ${10} # Script name & positional arguments
$#                      # Number of positional arguments
"$*"                   # All positional arguments as a single string
$?                     # Exit status of the last command
```

Program Flow Control

```
if [ condition ]; then command; fi          # Basic if statement
if [ condition ]; then command; else other_command; fi      # If-else statement
if [ condition ]; then command; elif [ condition ]; then... # If-elif-else statement
```

Test Conditions

```
# Numeric comparisons: -eq, -ne, -lt, -le, -gt, -ge
# File checks: -s, -f, -d, -x, -w, -r
```

```
# String comparisons: =, !=, -n (not zero), -z (is zero)
# Logical operators: && (and), || (or)
```

Loops and Functions

```
for i in {1..5}; do echo "Loop $i"; done      # For loop
while [ condition ]; do command; done        # While loop
case "$variable" in pattern) command;; esac    # Case statement
function name() { command; }                 # Function definition
name() { command; }                           # Alternative function syntax
name                                           # Call a function
```

Command Examples

```
crontab -e    # Edit crontab file
crontab -l    # List crontab entries
crontab -r    # Remove crontab entries
```